

ROUTING CONFIGURATION

Routing

- The Router (RTR) component uses database configuration to determine how a transaction should be routed to an authorizing entity.
- Transactions can be authorized on the BASE24-eps system by executing a script.
- Transactions can be authorized external from the BASE24-eps system.
- Once all configured routing options have been exhausted, transactions are authorized using a default authorization action.



Routing

- The Router component can invoke:
 - The Issuer Auth (IAUT) component for transactions requiring pre-screening or internal authorization services.
 - An external Issuer Interface component for transactions requiring external authorization services.
 - The Default Authorization component to authorize the transaction.

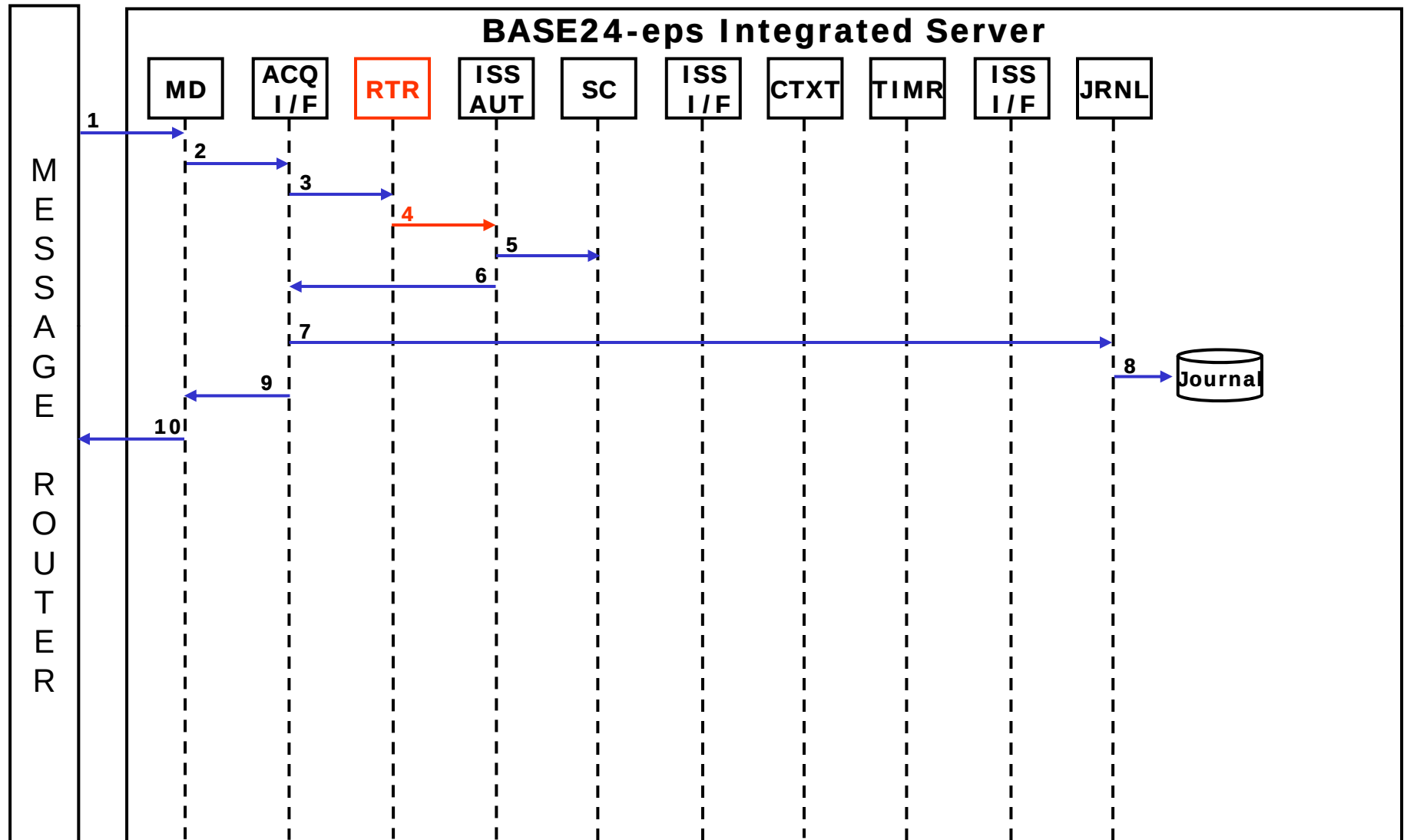


Routing Configuration

- When configuring routing destinations, the external authorizers (Interface records) must already be configured and available in the routing configuration
- If scripts are used for authentication (pre-screening) or authorization, they must also be configured and available in the routing configuration.

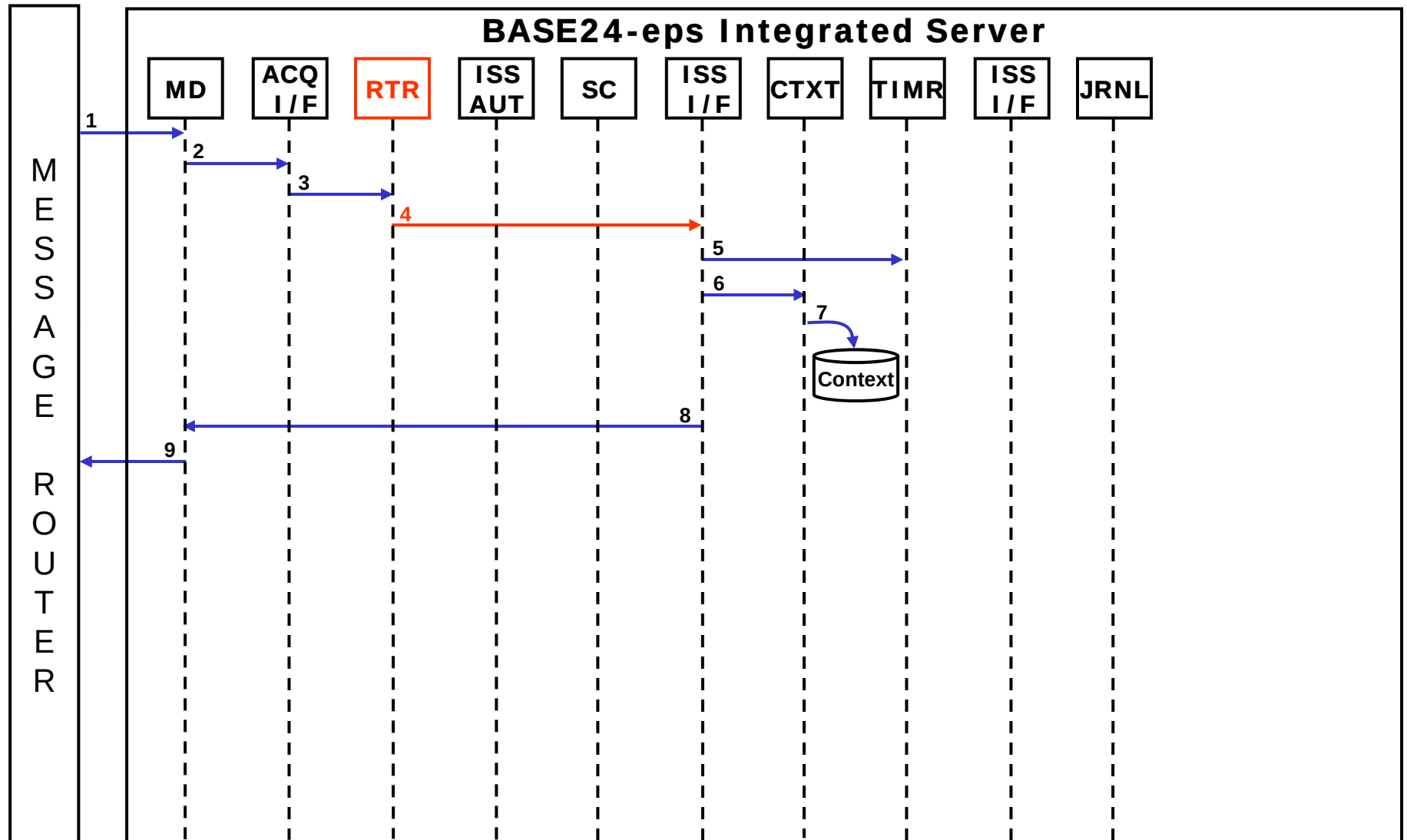


Component Interactions



Internal Authorization – Request/Response

Common Component Interactions



External Authorization – Request

Routing

- The payment instrument issuer determines how the transaction will be routed for authorization.
- The issuer may allow the transaction acquirer to have input to determine how the transaction is authorized.
- The Route Type configuration options include:
 - Acquirer and Issuer
 - Issuer Only
- If the Route Type is set to *Acquirer and Issuer*, then the source route profile configuration will also be used in determining the routing options.



Routing

- The Router component may use the following information to determine how a transaction should be routed:
 - route type
 - source route profile
 - destination route profile
 - transaction code
 - route code
 - account type 1 & 2
 - customer authentication method





ROUTING CONFIGURATION

Routing Configuration

Issuer and Acquirer Routing

The image displays two screenshots of ACI configuration windows. The top window, titled "On-Us Prefix Configuration - 160107 - 16", shows the "Route Type" dropdown menu with two options: "Acquirer and Issuer" (selected) and "Issuer only". The bottom window, titled "System Prefix Configuration - STAR_SRC", shows a table with columns for "Destination Routing Profile", "Issuer ID", "Limit Profile", "Issuer Surcharge Profile", "Instrument Type", and "Route Type". The "Route Type" dropdown menu is also shown, with "Acquirer & Issuer" and "Issuer only" options.

On-Us Prefix Configuration - 160107 - 16

Prefix: 160107 PAN Length: 16

Issuer ID: PCBA_INST (POS_CARD INSTITUTION)

Card Verification | Dynamic Card Verification | EMV General Options | EMV Auth Options | CAVV

Processing Information | Check Processing | Issuer Information | Card Track Information | PIN Verification

☐ Member Number on Track

☐ Include Holds When Checking Limits, Usages and Balances

Expiration Date Check Option: {none}

Expiration Date Comparison Option: No expiration date comparison required

Fraud System Option: Do Not Use Fraud System

Default Preauthorization Hold:

Route Type

- ☒ Acquirer and Issuer
- ☐ Issuer only

System Prefix Configuration - STAR_SRC

Source Routing Profile: STAR_SRC

Destination Routing Profile	Issuer ID	Limit Profile	Issuer Surcharge Profile	Instrument Type	Route Type
					Acquirer & Issuer
					Issuer only

Based on the Route Type configuration in the On-us Prefix data source or System Prefix data source, the Issuer can allow (configure) the Acquirer of a transaction to influence the routing of a transaction.

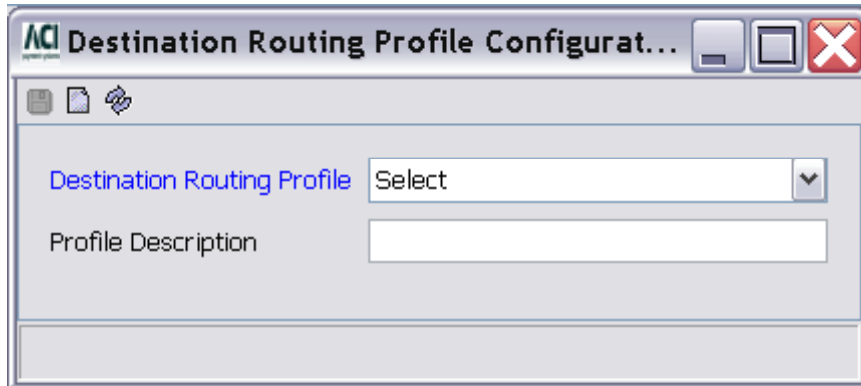
Destination/Source Routing Profile

The screenshot displays the ACI EPS Development - ESNCSERVICE application interface. The main menu on the left lists various configuration options, with 'Routing' selected. A sub-menu is open for 'Routing', showing options: Profile Description, Destinations, Destination/Source Relationship, and File Update Router. The 'Destination/Source Relationship' option is highlighted. Two configuration windows are overlaid on the main interface:

- Destination Routing Profile Configuration**: This window has a 'Destination Routing Profile' dropdown menu set to 'Select' and a 'Profile Description' text field. To its right, the text 'Select ONLY option' is displayed.
- Source Routing Profile Configuration**: This window has a 'Source Routing Profile' dropdown menu set to 'Select/Enter' and a 'Profile Description' text field. To its right, the text 'Select and Enter option' is displayed.

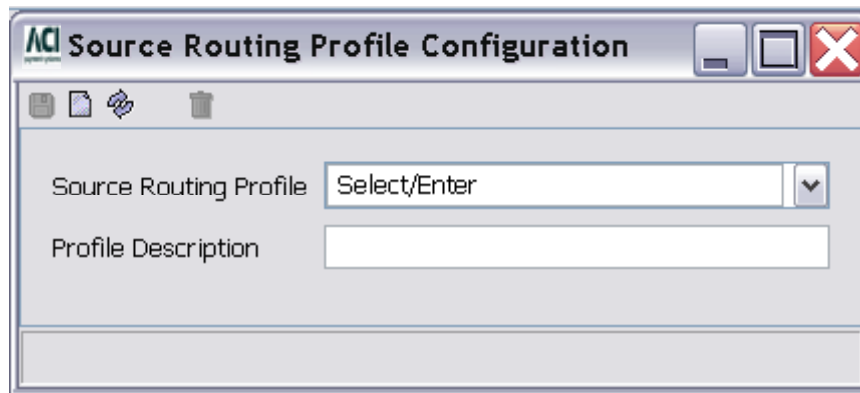
The status bar at the bottom indicates '277 - Security Information - Last successful login: 2/16/12 10:03 AM'.

Configuring Profile Values



The image shows a software window titled "Destination Routing Profile Configurat...". It features a toolbar with icons for save, print, and help. Below the toolbar, there is a label "Destination Routing Profile" followed by a dropdown menu currently showing "Select". Underneath this is a label "Profile Description" followed by a text input field. At the bottom of the window is a large empty rectangular area.

Destination Routing Profile values can be selected from the dropdown but NOT configured here.

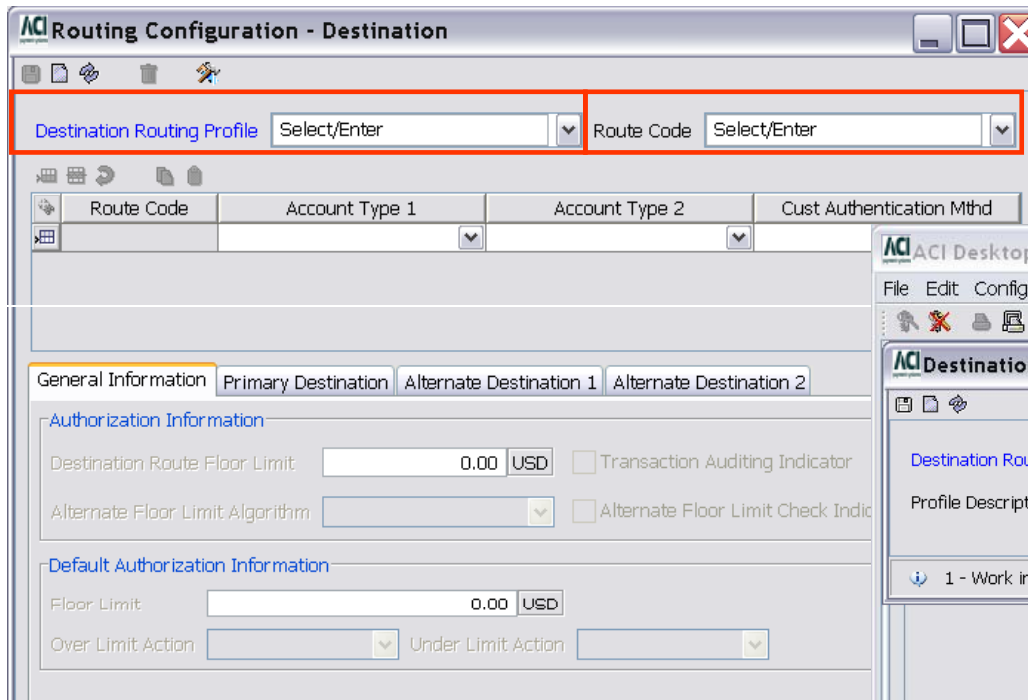


The image shows a software window titled "Source Routing Profile Configuration". It features a toolbar with icons for save, print, help, and delete. Below the toolbar, there is a label "Source Routing Profile" followed by a dropdown menu currently showing "Select/Enter". Underneath this is a label "Profile Description" followed by a text input field. At the bottom of the window is a large empty rectangular area.

Source Routing Profile values can be selected from the dropdown or configured here.

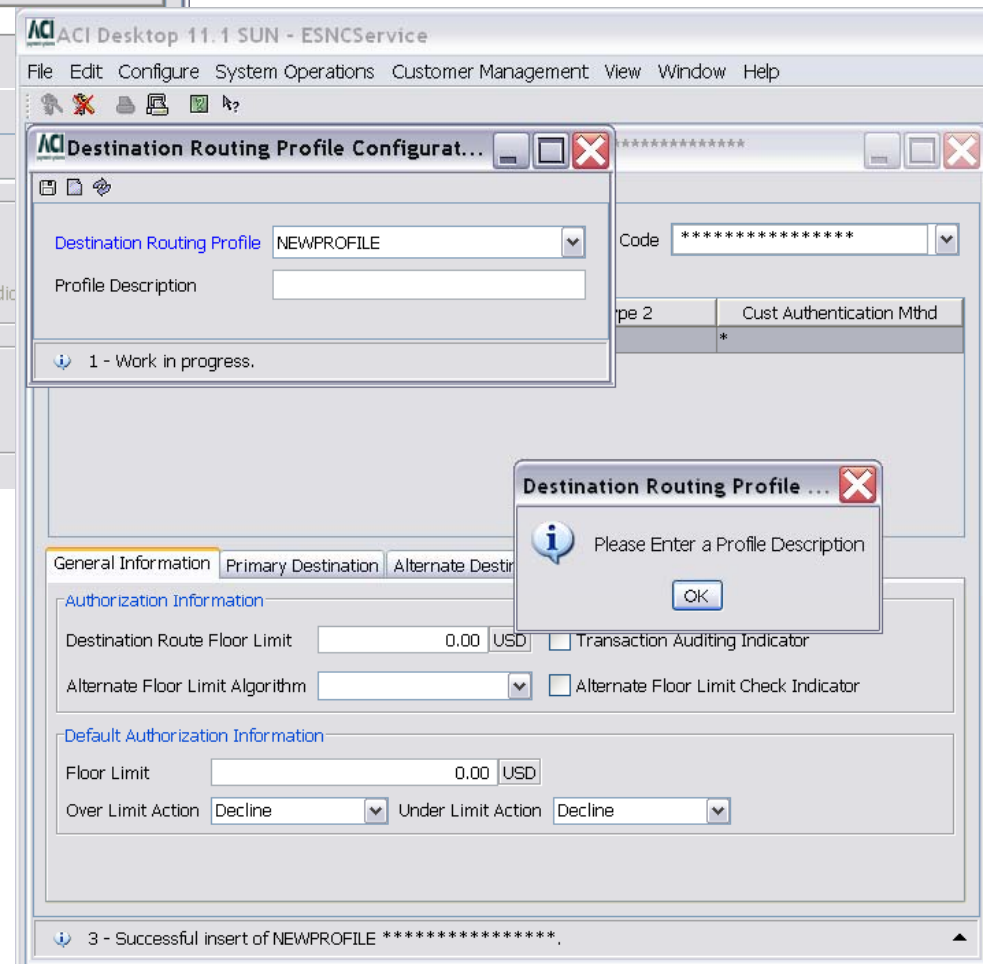
Destination Routing Profile

Destination Routing Profile values are configured in the Routing Configuration UI, as is the Route Code.



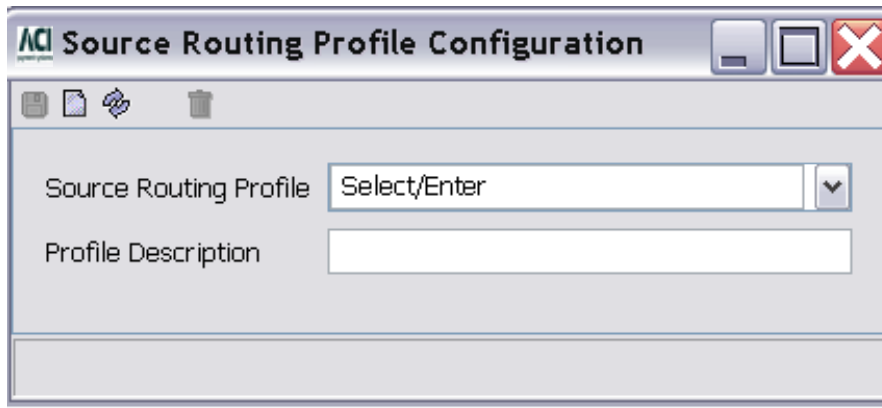
The screenshot shows the 'Routing Configuration - Destination' window. At the top, there are two dropdown menus: 'Destination Routing Profile' and 'Route Code', both with 'Select/Enter' as the current selection. Below these are tabs for 'Route Code', 'Account Type 1', 'Account Type 2', and 'Cust Authentication Mthd'. The 'General Information' tab is active, showing fields for 'Destination Route Floor Limit' (0.00 USD), 'Alternate Floor Limit Algorithm', and 'Default Authorization Information' (Floor Limit: 0.00 USD, Over Limit Action, Under Limit Action).

When the Routing Configuration record is added, a dialog box will display prompting to add a description for the Destination Routing Profile value. Then, the profile record is added to the Destination Routing Profile data source.



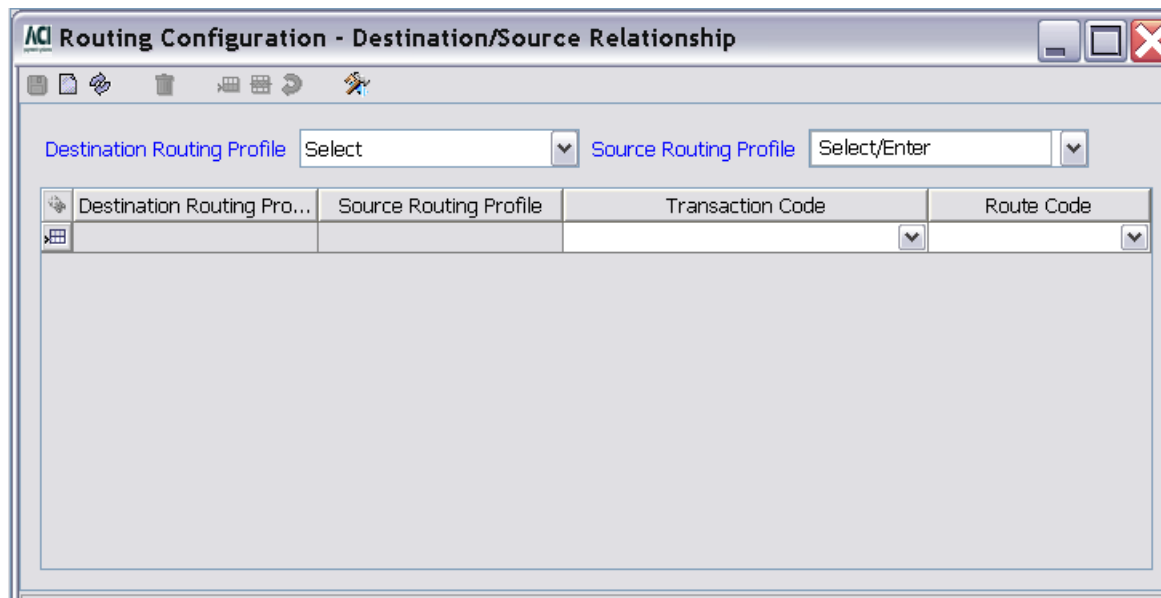
The screenshot shows the 'Destination Routing Profile Configuration' dialog box. It has a 'Destination Routing Profile' dropdown set to 'NEWPROFILE' and a 'Profile Description' field. Below this is a status bar showing '1 - Work in progress.' A smaller dialog box titled 'Destination Routing Profile ...' is overlaid, displaying an information icon and the message 'Please Enter a Profile Description' with an 'OK' button. The background window shows the same configuration fields as the first screenshot, but with the 'Over Limit Action' and 'Under Limit Action' dropdowns set to 'Decline'. The status bar at the bottom of the main window shows '3 - Successful insert of NEWPROFILE *****'.

Source Routing Profile



The image shows a window titled "Source Routing Profile Configuration". It contains two main input fields: "Source Routing Profile" with a dropdown menu showing "Select/Enter" and a small downward arrow, and "Profile Description" with a text input field. There are also icons for saving, deleting, and other actions at the top left of the window.

Source Routing Profile values can be selected or entered in the Source Routing Profile Configuration UI.

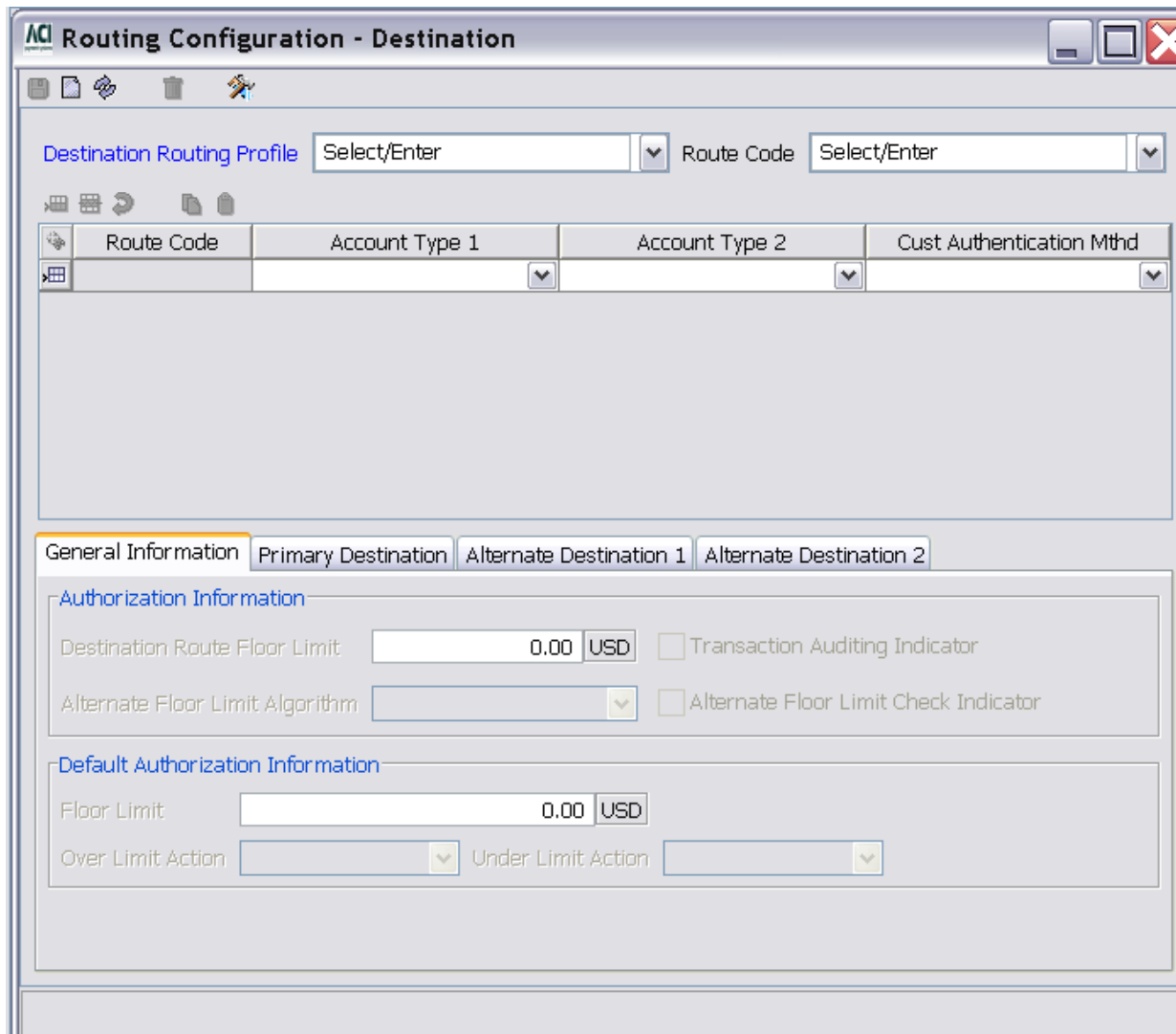


The image shows a window titled "Routing Configuration - Destination/Source Relationship". It features two dropdown menus at the top: "Destination Routing Profile" with "Select" and "Source Routing Profile" with "Select/Enter". Below these is a table with four columns: "Destination Routing Pro...", "Source Routing Profile", "Transaction Code", and "Route Code". The table has a header row and one data row with dropdown arrows in the last two columns. There are also icons for saving, deleting, and other actions at the top left of the window.

Destination Routing Pro...	Source Routing Profile	Transaction Code	Route Code

Source Routing Profile values can be selected or entered here in the Routing Configuration – Destination/Source Relationship UI.

Routing Configuration - Destination



ACI Routing Configuration - Destination

Destination Routing Profile: Select/Enter Route Code: Select/Enter

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

Authorization Information

Destination Route Floor Limit: 0.00 USD ☐ Transaction Auditing Indicator

Alternate Floor Limit Algorithm: ☐ Alternate Floor Limit Check Indicator

Default Authorization Information

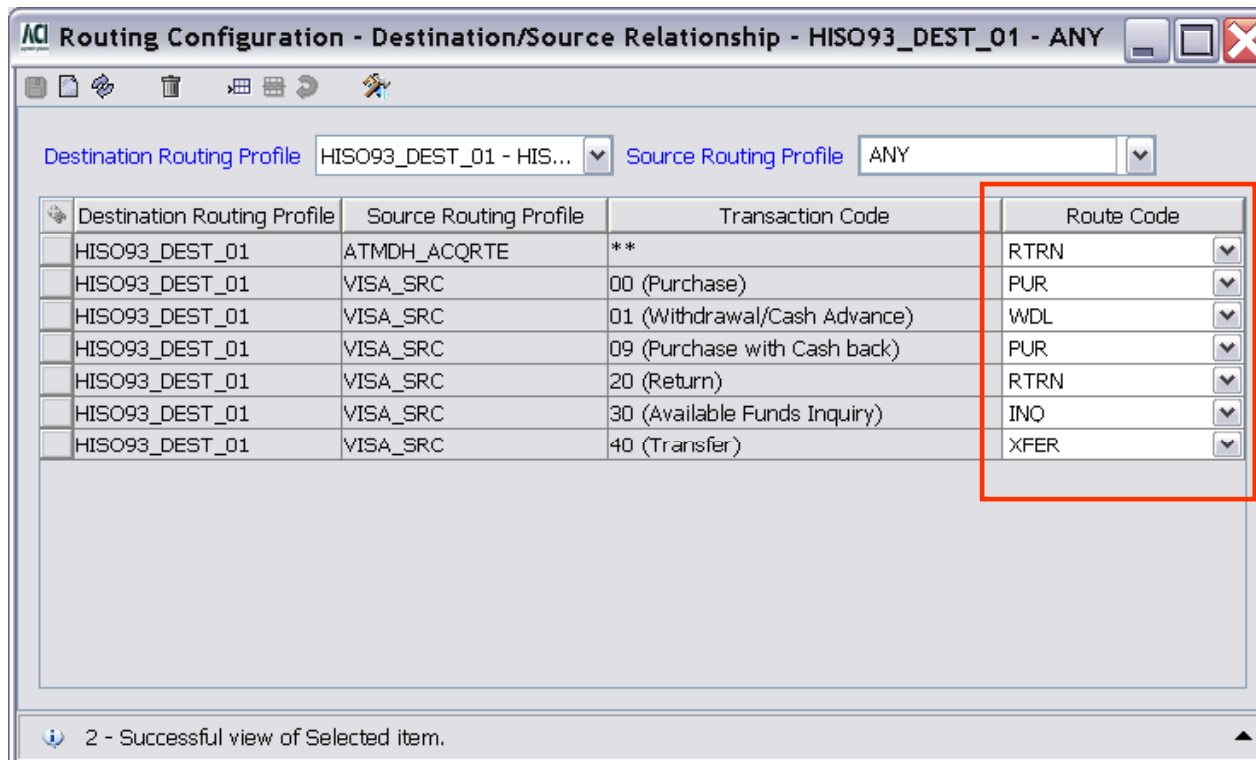
Floor Limit: 0.00 USD

Over Limit Action: Under Limit Action:

Only on the Routing Configuration UI can you configure the Route Code.

Once defined here, the Route Code value will populate the drop down list available in the Routing Configuration – Destination/Source Relationship UI.

Routing Configuration – Destination/Source Relationship UI



Destination Routing Profile	Source Routing Profile	Transaction Code	Route Code
HISO93_DEST_01	ATMDH_ACQRTE	**	RTRN
HISO93_DEST_01	VISA_SRC	00 (Purchase)	PUR
HISO93_DEST_01	VISA_SRC	01 (Withdrawal/Cash Advance)	WDL
HISO93_DEST_01	VISA_SRC	09 (Purchase with Cash back)	PUR
HISO93_DEST_01	VISA_SRC	20 (Return)	RTRN
HISO93_DEST_01	VISA_SRC	30 (Available Funds Inquiry)	INO
HISO93_DEST_01	VISA_SRC	40 (Transfer)	XFER

Route Codes are available from the drop down on the Destination/Source Relationship UI.

This data source is only used if the Acquirer/Issuer route type is configured.

Determining Routing Configuration

1. Become familiar with the network routing requirements and lay out a plan. Determine if Acquirer/Issuer routing will be implemented.
 - Is it necessary to take the transaction acquirer into account for special routing considerations?
2. Determine the Destination/Source relationships
 - Layout the Destination and Source profile names
 - Establish the Route Code name that represents the Destination/Source/Transaction Code relationships required.
3. Add the Source profile names into the Source Routing Profile Configuration UI.
4. Configure the Routing Configuration-Destination UI
 - Enter the Destination Routing profile name and the Route Code name.

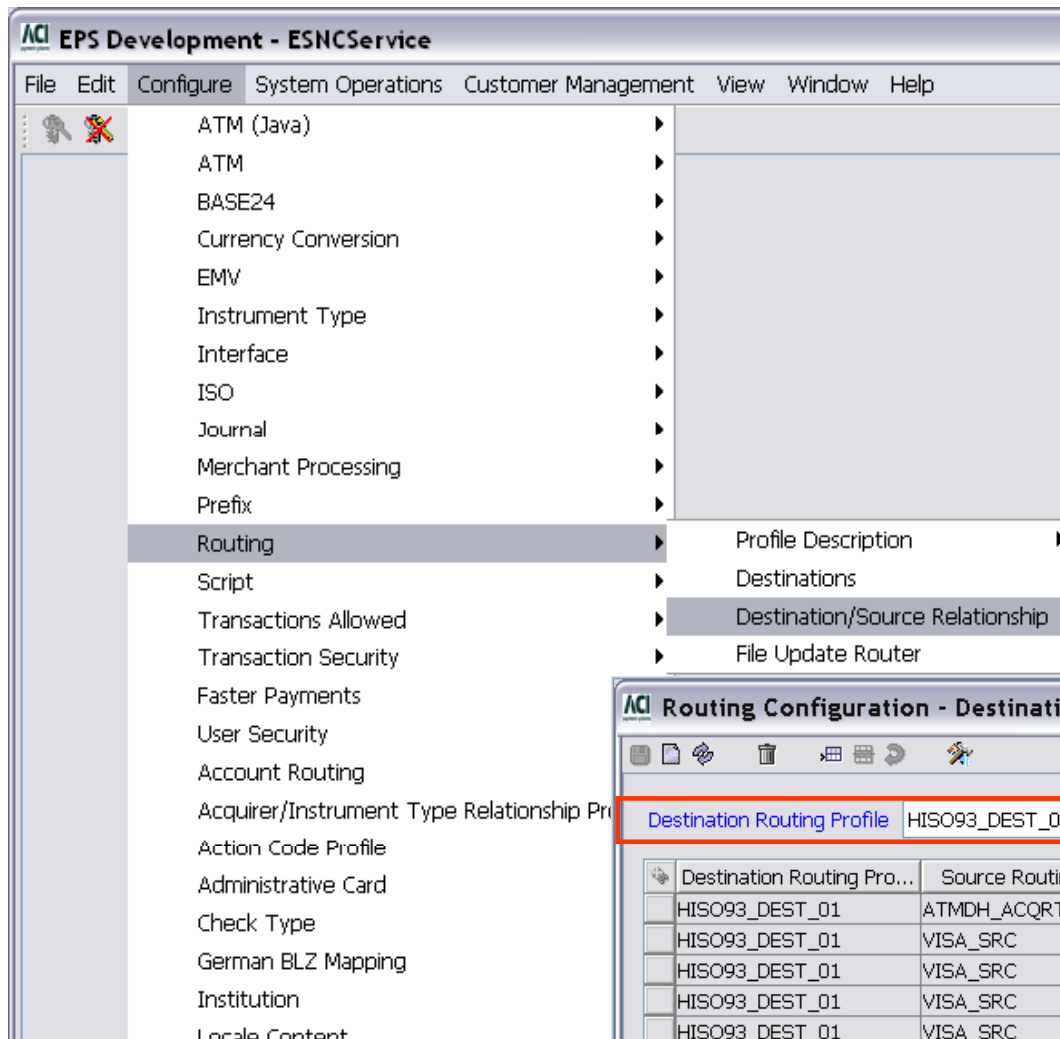


Determining Routing Configuration

5. Select the appropriate Destination/Route Code combination
 - Configure the Primary, Alternate 1, Alternate 2, and Default routing actions.
6. Add the Routing Configuration record.
 - A dialogue box will prompt the user to add the Destination Profile Name to the data source.
7. Configure the Destination Routing Profile, Source Routing Profile, and Transaction Code relationship, if necessary, in the Routing Configuration – Destination/Source Relationship UI.

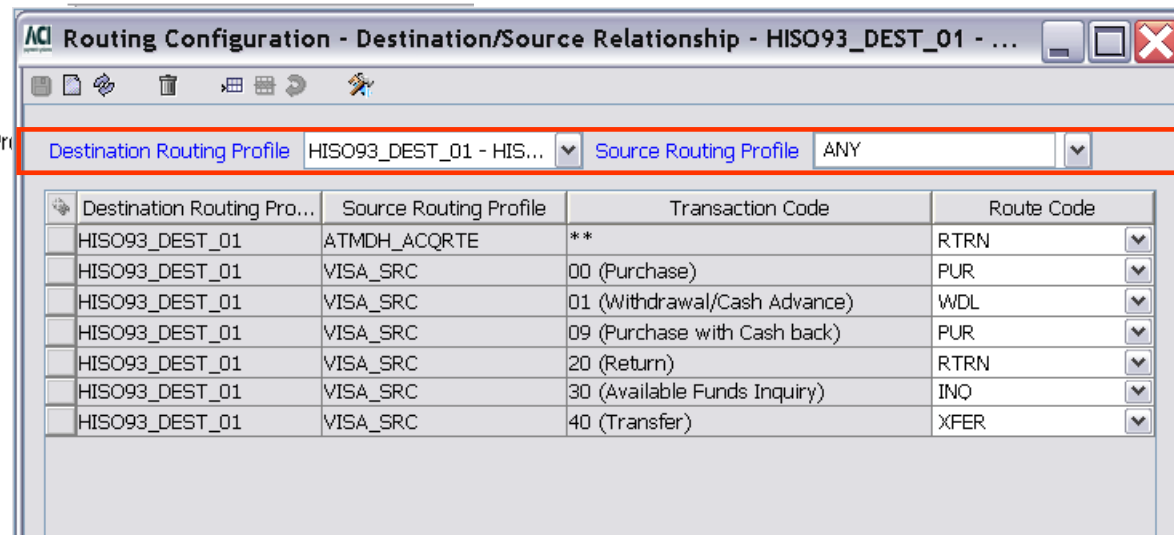


Destination/Source Relationship

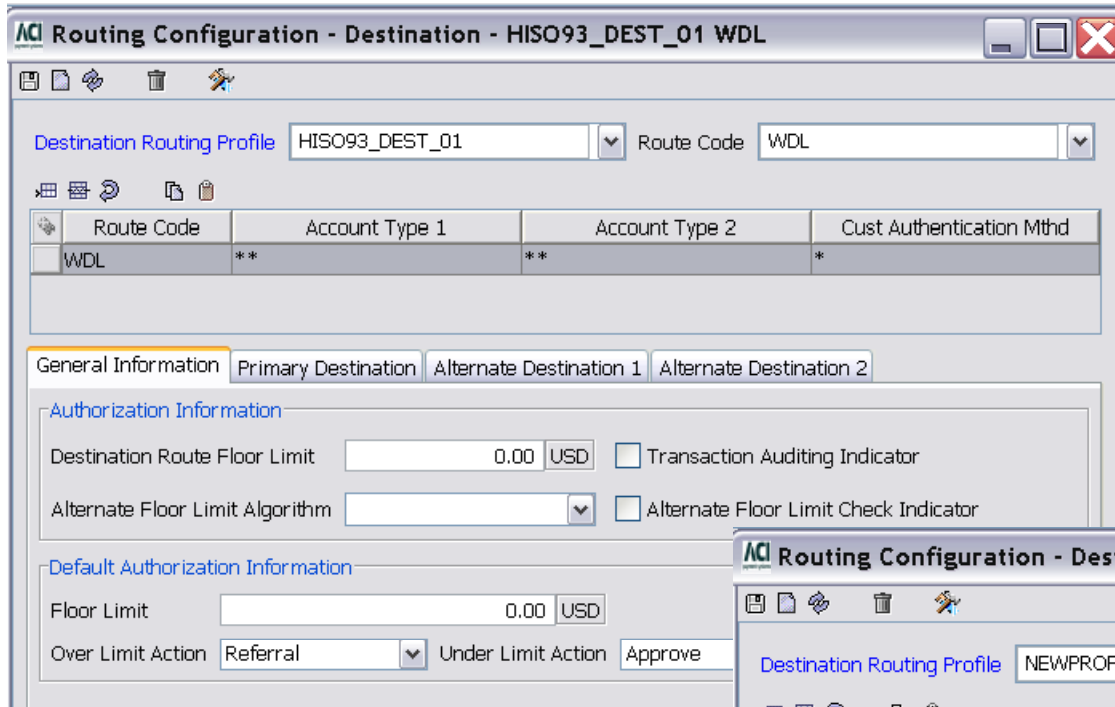


The Acquirer/Issuer Relation data source will be used if Route Type is set to *Acquirer and Issuer*.

If Route type is set to *Issuer Only*, this data source is not used.



Route Code



Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

Authorization Information

Destination Route Floor Limit: 0.00 USD ☐ Transaction Auditing Indicator

Alternate Floor Limit Algorithm: ☐ Alternate Floor Limit Check Indicator

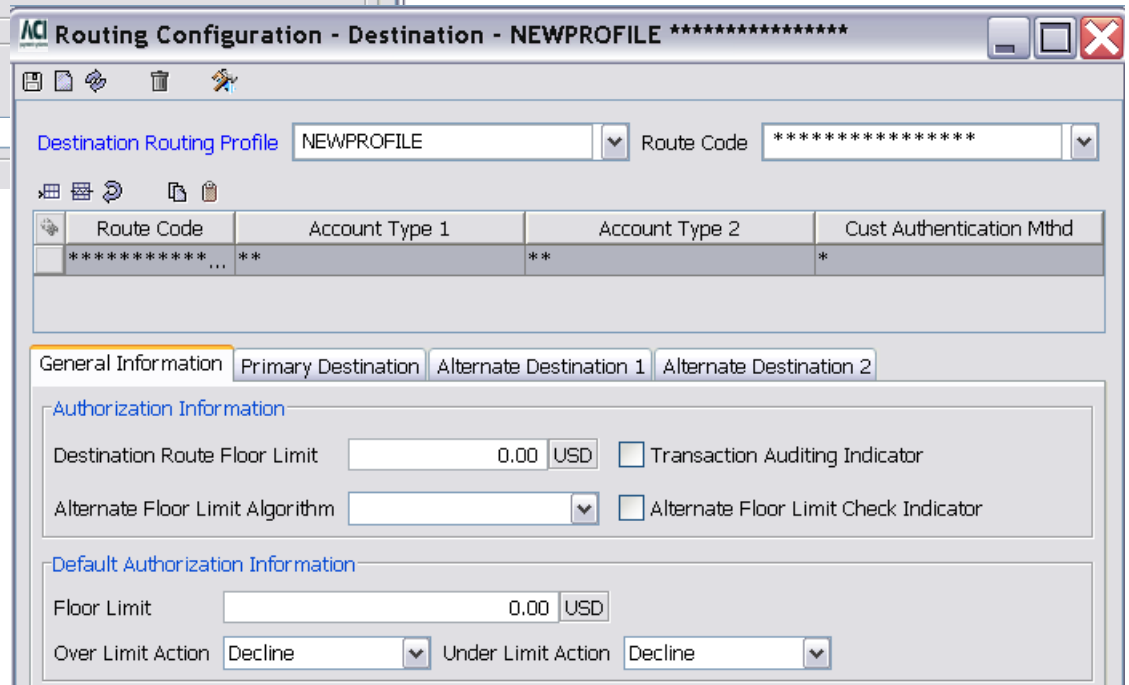
Default Authorization Information

Floor Limit: 0.00 USD

Over Limit Action: Referral Under Limit Action: Approve

A Route Code derived from the Acquirer/Issuer Relation data source allows for configuration of unique routing destinations based on the Transaction Code and the relationship between the Acquirer and Issuer.

A Route Code of all asterisks will be used if transaction routing is only based on the Issuer's requirements.



Routing Configuration - Destination - NEWPROFILE *****

Destination Routing Profile: NEWPROFILE Route Code: *****

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
*****	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

Authorization Information

Destination Route Floor Limit: 0.00 USD ☐ Transaction Auditing Indicator

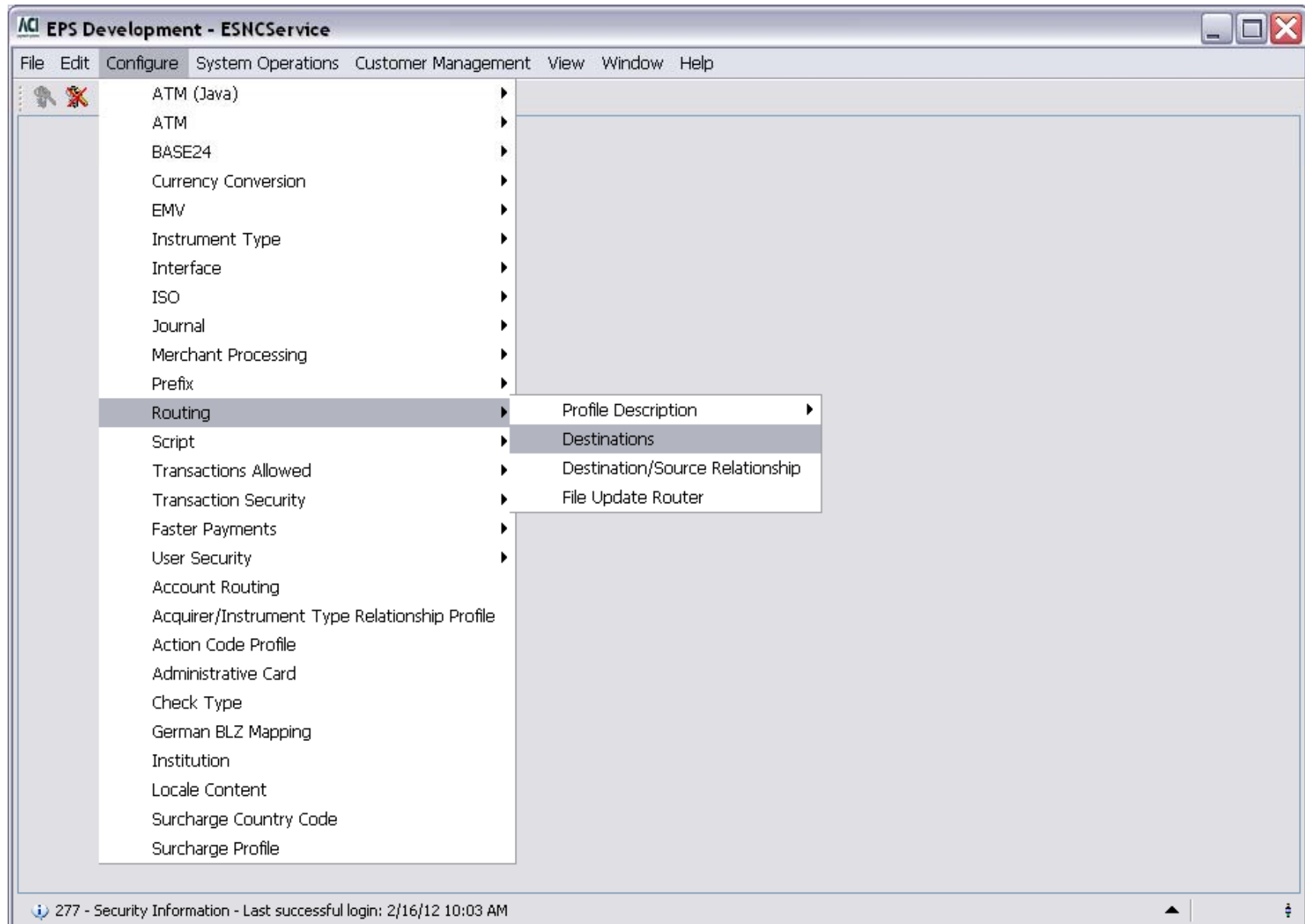
Alternate Floor Limit Algorithm: ☐ Alternate Floor Limit Check Indicator

Default Authorization Information

Floor Limit: 0.00 USD

Over Limit Action: Decline Under Limit Action: Decline

Routing Configuration



Routing Configuration

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ACI Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

Authorization Information

Destination Route Floor Limit: 0.00 USD ☐ Transaction Auditing Indicator

Alternate Floor Limit Algorithm: ☐ Alternate Floor Limit Check Indicator

Default Authorization Information

Floor Limit: 0.00 USD

Over Limit Action: Referral Under Limit Action: Approve

2 - Successful view of HISO93_DEST_01 WDL.

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Routing Configuration

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Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

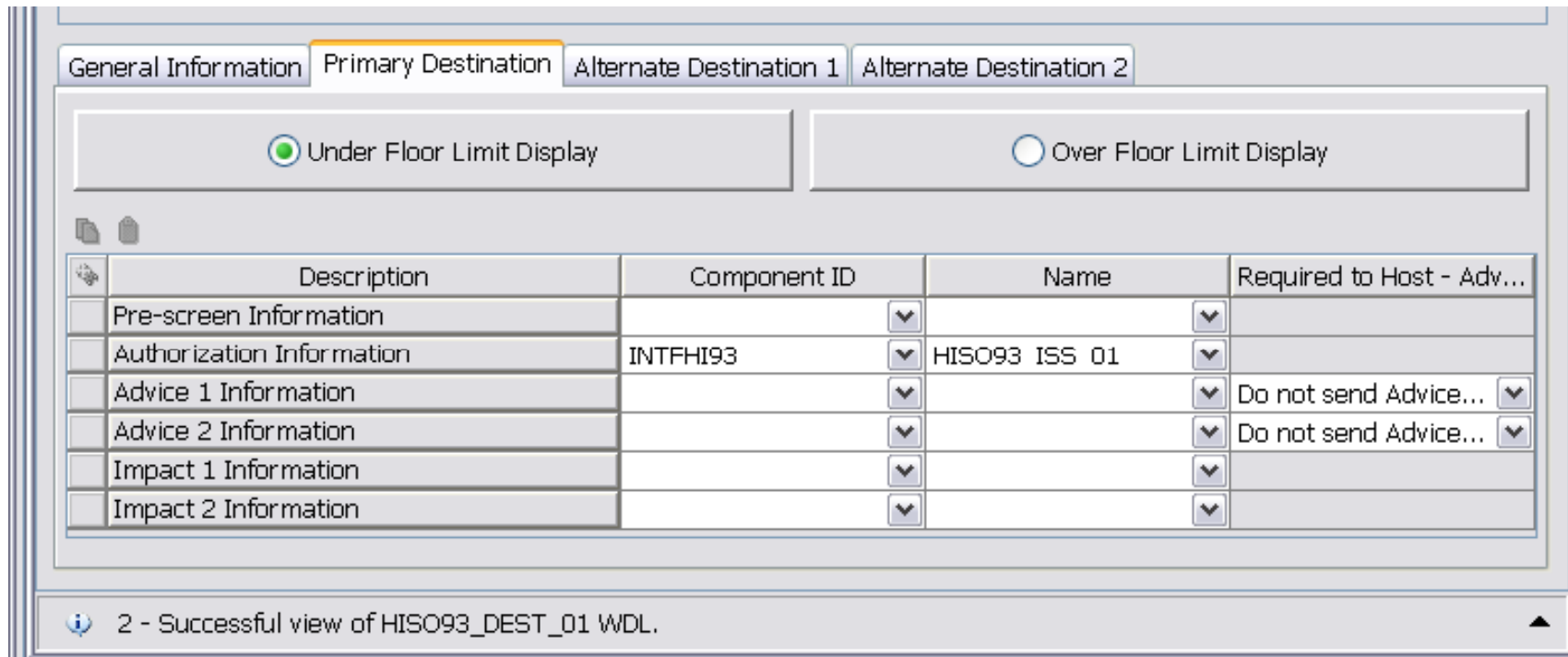
☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Advice ...
Pre-screen Information			
Authorization Information	INTFHI93	HISO93_ISS_01	
Advice 1 Information			Do not send Advice m... ☐
Advice 2 Information			Do not send Advice m... ☐
Impact 1 Information			
Impact 2 Information			

2 - Successful view of HISO93_DEST_01 WDL.

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Routing Configuration



General Information Primary Destination Alternate Destination 1 Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Adv...
Pre-screen Information			
Authorization Information	INTFHI93	HISO93 ISS 01	
Advice 1 Information			Do not send Advice... v
Advice 2 Information			Do not send Advice... v
Impact 1 Information			
Impact 2 Information			

2 - Successful view of HISO93_DEST_01 WDL.

- Description
 - Pre-screen Information: configures a pre-screening script to execute
 - Authorization Information: configures the transaction authorization destination
 - Advice 1 (Advice 2) Information: configures advice message actions
 - Impact 1 (Impact 2) Information: configures database update scripts

Routing Configuration

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Adv...
☐ Pre-screen Information			
☐ Authorization Information	INTFHI93	HISO93 ISS 01	
☐ Advice 1 Information	INTFHI93		Do not send Advice...
☐ Advice 2 Information	INTFINTR		Do not send Advice...
☐ Impact 1 Information	INTFIPAY		
☐ Impact 2 Information	INTFJENI		
	INTFMDS		
	INTFPULS		
	INTFRTEF		

- Component:
 - A selectable list of Issuer components
 - Includes Issuer Auth and each Issuer Interface component

Routing Configuration

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Adv...
Pre-screen Information			
Authorization Information	IAUT	NEGU_DEP	
Advice 1 Information		NEGU_DEP	do not send Advice...
Advice 2 Information		NEGU_PUR	do not send Advice...
Impact 1 Information		NEGU_PUR_IMP	
Impact 2 Information		NEGU_PUR_PS	
		NEGU_WDL	
		NEGU_WDL_IMP	
		NEGU_WDL_PS	
		NEGU_WDL_YEP	

If IAUT component is selected, the NAME dropdown will include all sets of authorization scripts configured.

Authorization Script Configuration - NEGU_WDL

Authorization Name: NEGU_WDL

Message Type	Script Name	Script Enable Flag	Monitor Enable Flag	Approval Threshold	Referral Threshold	Denial Threshold	Minimum Transaction Count
Authorization Request	NEGU_WDL_AU_RQ	☒	☒	0	99	99	300
Authorization Advice	NEGU_WDL_AU_FP	☒	☒	0	99	99	300
Financial Request	NEGU_WDL_AU_RQ	☒	☒	0	99	99	300
Financial Advice	NEGU_WDL_AU_FP	☒	☒	0	99	99	300
Reversal Advice	NEGU_WDL_AU_RV	☒	☒	0	99	99	300

2 - Successful view of NEGU_WDL.

Routing Configuration

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - A...
Pre-screen Information			
Authorization Information	INTFHI93	HISO93 ISS_01	
Advice 1 Information		HISO93	Do not send Advi...
Advice 2 Information		HISO93_ISS_01	Do not send Advi...
Impact 1 Information		HISO93_ISS_02	
Impact 2 Information			

If an Interface component is selected, the NAME dropdown will include all selections configured for that type of Interface.

ACI ISO8583 (93) Host Interface Configuration

Interface Name

Interface Cutover Time Time Offset From Processor (hrs:mins)

Statistics Update Interval

Settlement Processing Days

☐ 5-Day (Business Day) ☒ 7-Day

Interface Profiles

Source Routing Profile

Acquirer Transactions Allowed Profile

Issuer Transactions Allowed Profile

Context Profile

ISO Message Profile

Action Code Profile

Acquirer Surcharge Profile

Currency Conversion Fee Profile

Routing Configuration

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ACI Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Advice ...
Pre-screen Information			
Authorization Information	INTFHI93	HISO93_ISS_01	
Advice 1 Information			Do not send Advice m... ☐
Advice 2 Information			Do not send Advice m... ☐
Impact 1 Information			
Impact 2 Information			

2 - Successful view of HISO93_DEST_01 WDL.

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Routing Configuration

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ACI Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination **Alternate Destination 1** Alternate Destination 2

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Advice ...
Pre-screen Information			
Authorization Information			
Advice 1 Information			Do not send Advice m...
Advice 2 Information			Do not send Advice m...
Impact 1 Information			
Impact 2 Information			

2 - Successful view of HISO93_DEST_01 WDL.

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Routing Configuration

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ACI Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 **Alternate Destination 2**

☒ Under Floor Limit Display ☐ Over Floor Limit Display

Description	Component ID	Name	Required to Host - Advice ...
Pre-screen Information			
Authorization Information			
Advice 1 Information			Do not send Advice m...
Advice 2 Information			Do not send Advice m...
Impact 1 Information			
Impact 2 Information			

2 - Successful view of HISO93_DEST_01 WDL.

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Default Authorization Configuration

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ACI Routing Configuration - Destination - HISO93_DEST_01 WDL

Destination Routing Profile: HISO93_DEST_01 Route Code: WDL

Route Code	Account Type 1	Account Type 2	Cust Authentication Mthd
WDL	**	**	*

General Information Primary Destination Alternate Destination 1 Alternate Destination 2

Authorization Information

Destination Route Floor Limit: 0.00 USD ☐ Transaction Auditing Indicator

Alternate Floor Limit Algorithm: ☐ Alternate Floor Limit Check Indicator

Default Authorization Information

Floor Limit: 0.00 USD

Over Limit Action: Referral Under Limit Action: Approve

2 - Successful view of HISO93_DEST_01 WDL.

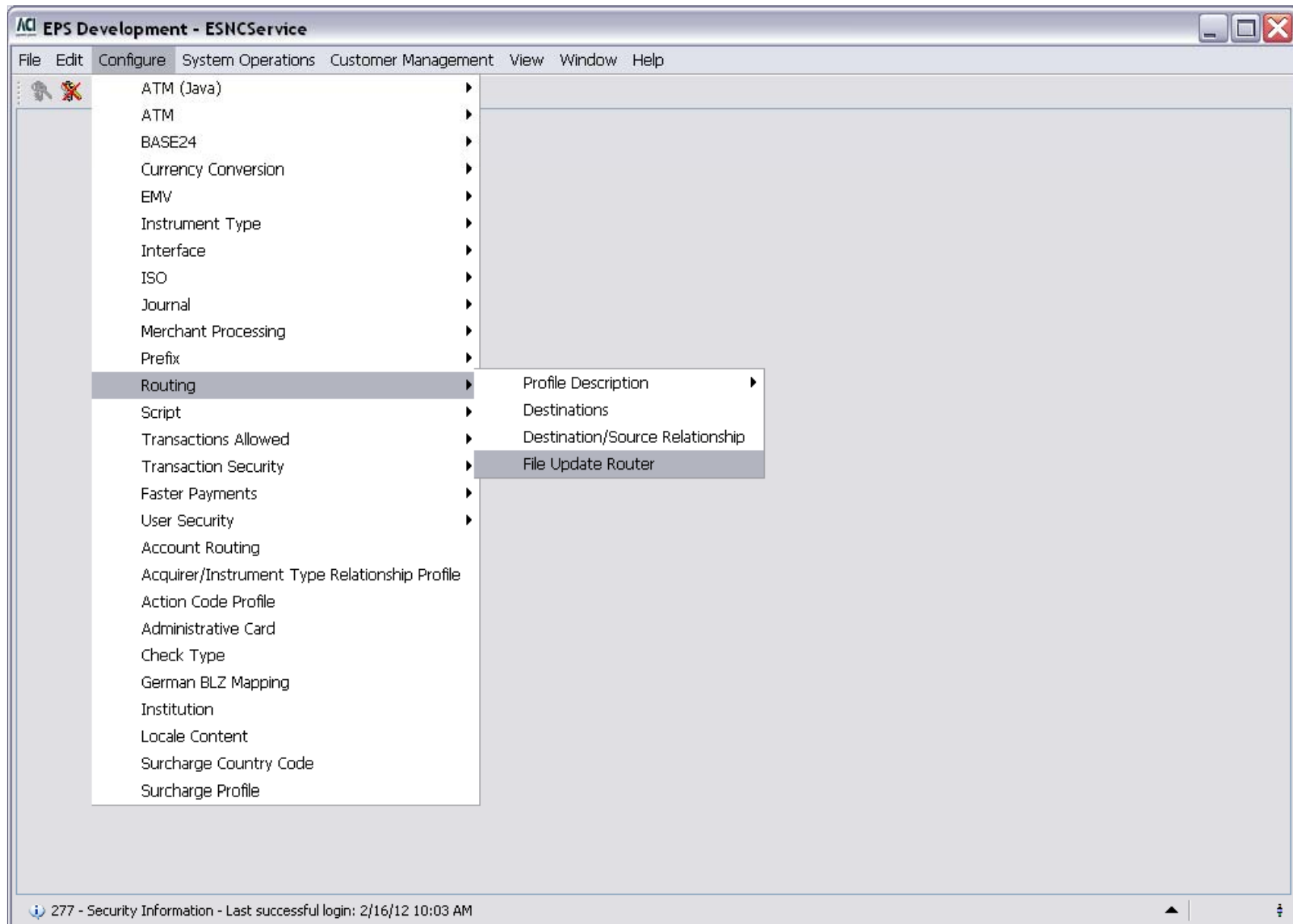
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File Update Router

- Routes data source update notifications that are:
 - received from external entities (i.e., a host or interchange) to be applied to the BASE24-eps database and/or forwarded to an external entity
 - generated within a script and sent to external entities
- Updates to the Card data source in script processing can be routed to external entities if the script invokes CARD processing using a “_NOTIFY” exported operator.
- Updates could include Visa Card Recovery Bulletin, Banknet Account and Account Management Files, STAR Negative File, etc., CARD data source and Positive Balance data source.



File Update Router Configuration



File Update Router Configuration

ACI EPS Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI File Update Router Configuration - HISO93_DEST_01 and ATMDH_ACQRTE

Destination Routing Profile: HISO93_DEST_01 Source Routing Profile: ATMDH_ACQRTE

	Source Routing Profile	File Type	Component ID	Interface Name
+	ATMDH_ACQRTE	Card Table (01)	INTFSTAR	DH_STAR
+	ATMDH_ACQRTE	Positive Balance (02)	INTFMDS	MDS
+	ATMDH_ACQRTE			

277 - Security Information - Last successful login: 2/16/12 10:03 AM

Routing Configuration

- Changes to the Routing configuration table(s) requires rebuilding and reloading the respective OLTP table(s):
 - acquirer_issuer_relation_oltp
 - route_oltp
 - file_update_route_oltp
- OLTP tables can be built and loaded using Process Control or the OLTP Warmboot Management UI.



Lab Exercise

BASE24-eps Routing Configuration Lab

Lab 3





MEETS THE CHALLENGE OF CHANGE